Dog Breed Identification using Transfer Learning

Opening

Hello everyone, welcome to this project demonstration on Dog Breed Identification using Transfer Learning. This video explains the objective, dataset, model, and results of the project.

Project Overview

Dog breed identification is an image classification task. This project uses transfer learning with a pre-trained deep learning model to accurately classify different dog breeds from images.

Objectives

• Identify different dog breeds from images
• Improve accuracy using transfer learning
• Reduce training time
• Build an efficient image classification model

Dataset Description

The dataset contains labeled images of different dog breeds collected from public datasets like Kaggle or ImageNet.

Data Preprocessing

Images are resized, normalized, and augmented for better model performance.

Model Used

Pre-trained models like ResNet50, VGG16, or MobileNet are used. The base layers are frozen and new classification layers are added.

Training and Testing

The model is trained on the dataset and tested for accuracy and performance.

Key Results

• High classification accuracy
• Faster training time
• Accurate breed prediction with confidence score

Applications

Used in pet identification apps, veterinary systems, and animal rescue organizations.

Conclusion

Transfer learning provides an efficient and accurate solution for dog breed classification.

Closing

Thank you for watching. The complete project is available in the GitHub repository.